

2(a)(i)	NADP ;	1
2(a)(ii)	<i>two from:</i> 1 photolysis of water / AW ; 2 <i>ref. to</i> photosystem II ; 3 using an enzyme ; 4 hydrogen ions and electrons combine to form hydrogen / AW ;	2

Question	Answer	Marks
2(b)(i)	<i>two from:</i> 1 to, reduce / stop, activity of enzymes ; 2 <i>ref. to</i> proteases / lipases ; 3 to prevent damage (to chloroplasts) ;	2
2(b)(ii)	1 to, control pH / keep pH constant ; 2 so enzyme works at optimum / to prevent denaturation of enzyme ;	2
2(b)(iii)	1 to avoid, osmosis / movement of water down a water potential gradient ; 2 to prevent damage (to chloroplasts) / AW ;	2
2(c)(i)	43 / 43.0 ;	1
2(c)(ii)	23.3 / 23.0 / 23 ; <i>accept ecf from 2(c)(i)</i>	1

Question	Answer	Marks
2(c)(iii)	<i>four from:</i> <i>relationship</i> 1 as light intensity increases time taken for decolourisation decreases or as light intensity increases rate of decolourisation increases or as light intensity increases decolourisation is faster ; <i>explanation (max 3)</i> 2 more, photons / <u>light energy</u>, absorbed ; 3 more / faster, photolysis / AW ; 4 more electrons, excited / released or faster rate of electron release ; 5 more protons released or faster rate of proton release ; 6 more / faster, <u>reduction</u> of DCPIP ;	4